IGCSE 0607 Practice Questions — E3.3 — Combined Questions & Markscheme

Functions

Total Questions: 39 — **Total Marks:** 301 Marks

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- **E3.3** Functions
  → **Paper 2 (Non-Calculator)** *(22 Qs)* · **Paper 4 (Calculator)** *(17 Qs)*

1 Functions f and g are defined as follows.

$f(x) = 2x - 5$, $g(x) = x^2 + 1$

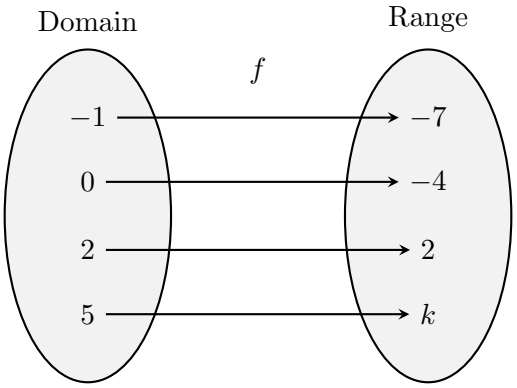
(a) Find the value of $f(g(3))$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

2 The mapping diagram shows the function $f(x) = 3x - 4$ for the domain $\{-1, 0, 2, 5\}$.



(a) Work out the value of k.

$k =$ ..... **[1]**

(b) Write down the range of f.

..... **[1]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

3 It is given that $f(x) = 3x - 2$ and $g(x) = \dfrac{10}{x}$.

(a) Find $f(4)$.

..... **[1]**

(b) Find x when $g(x) = 2$.

$x =$ ..... **[1]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

4 The functions f and g are defined as $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

(a) Work out

(i)

hspace*0.8cm $f(-4)$

..... **[1]**

(ii)

hspace*0.8cm $gf(2)$

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

5 $f(x) = 4x - 5$
$g(x) = \dfrac{x}{3} + 2$

(a) Work out $f(-2)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $fg(x)$ in its simplest form.

$fg(x) =$ ..... **[2]**

6 The functions f and g are defined as $f(x) = 5x - 2$, $g(x) = x^2 + 3$.

(a) Find the value of $f(4)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $fg(2)$.

..... **[2]**

7 The functions f and g are defined as $f(x) = 4x - 5$, $g(x) = \dfrac{12}{x}$.

(a) Work out the value of $f(3)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Work out $fg(2)$.

..... **[2]**

8 Consider the functions $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

(a) Find $f(4)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $gf(2)$.

..... **[2]**

9 Consider the functions $f(x) = 4x - 7$ and $g(x) = 2x^2 + 1$.

(a) Find $f(-3)$.

..... **[1]**

(b) Find $gf(2)$.

..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

10 Let $f(x) = 5 - 2x$, $g(x) = x^2 - 3$.

(a) Work out $f(-4)$.

..... **[1]**

(b) Work out $gf(2)$.

..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

11 $f(x) = 4x - 3$, $g(x) = 2 - 5x$

(a) Work out $f(2)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Work out $gf(1)$.

..... **[2]**

12 Let $f(x) = 4x - 3$ and $g(x) = 2x^2 + 1$.

(a) Work out $fg(3)$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Solve the equation

$$f(x) = 17$$

$x =$ ..... **[2]**

13 Two functions, f and g, are defined as follows:

$$f(x) = 3x - 2 \qquad g(x) = x^2 + 1$$

(a) Work out $f(4)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

14 Consider the functions $f(x) = \dfrac{5}{x-3}$ for $xe3$, and $g(x) = 2x + 1$.

(a) Find $fg(x)$, giving your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Solve the equation $gf(x) = 6$.

$x =$ ..... **[2]**

15 The functions f and g are defined as $f(x) = \dfrac{3x+2}{x-1}$, $x \neq 1$, $g(x) = x - 2$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[4]**

16 Consider the functions $h(x) = \dfrac{12}{x-1}$ and $k(x) = 2x + 3$.

(a) State the value of x which must be excluded from the domain of $h(x)$.

$x =$ ..... **[1]**

(b) Find $k^{-1}(x)$.

$k^{-1}(x) =$ ..... **[2]**

(c) Find $hk(x)$, giving your answer in its simplest form.

$hk(x) =$ ..... **[2]**

(d) Solve the equation.

$h(x) = k(2)$

$x =$ ..... **[3]**

17 Let $f(x) = \dfrac{4}{2x-3}$ for $x \neq 1.5$, and $g(x) = x + 5$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation $f(x) = g(-3)$.

$x =$ ..... **[3]**

(c) Express $gf(x)$ as a single fraction in its simplest form.

$gf(x) =$ ..... **[3]**

18 The functions f, g and h are defined as $f(x) = x^2 - 2$, $g(x) = \dfrac{x-3}{2}$, $h(x) = 5 - 3x$.

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

(b) Solve the equation

$f(x) = 34$

$x =$ ..... , $x =$ ..... **[2]**

(c) Find $hg(x)$ in its simplest form.

..... **[3]**

(d) Work out $fh(2)$.

..... **[2]**

19 The functions f and g are defined as $f(x) = \dfrac{x+3}{x-1}$, where $x \neq 1$, and $g(x) = 2x - 5$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Show that $ff(x) = x$.

[3]

(c) Solve the equation $g(x^2) = f(3)$.

$x =$ ..... **[4]**

20 $f(x) = \dfrac{3x+2}{x-1}$, $g(x) = \dfrac{2}{x+1}$

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Find $fg(x)$, giving your answer as a single fraction in its simplest form.

..... **[3]**

(c) Solve the equation.

$f(x) = f^{-1}(x)$

Give your answers in the form $a \pm \sqrt{b}$, where a and b are integers.

$x =$ ..... **[4]**

21 The functions f and g are defined as follows:

$f(x) = \dfrac{3x+5}{x-1}, x \neq 1$, and $g(x) = x^2 + 1$

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(b) Solve the equation $f(x) = x$.

$x =$ ..... , $x =$ ..... **[4]**

(c) Find $gf(2)$.

$gf(2) =$ ..... **[2]**

22 $f(x) = \dfrac{4}{x-1}$ for $xe1$, $g(x) = 2x + 3$, $h(x) = x^2 - 4$.

(a) Find an expression, in its simplest form, for

(i) $fg(x)$,

$fg(x) =$ ..... **[2]**

(ii) $fh(x) - 1$ as a single fraction.

..... **[3]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation $gf(x) = 5$.

$x =$ ..... **[3]**

23 The functions h and k are defined as $h(x) = \dfrac{2x+1}{3}$ and $k(x) = 5 - x$.

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

(b) Find an expression for $hk(x)$ in its simplest form.

$hk(x) =$ ..... **[2]**

(c) Solve the equation $h(x) = k(x)$.

$x =$ ..... **[3]**

24 Functions h and k are defined as follows.

$h(x) = \dfrac{4}{x-3}$ for $x \neq 3$

$k(x) = 2x + 1$

(a) Find $k(h(5))$.

..... **[2]**

(b) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

(c) Solve the equation $h(k(x)) = 2$.

$x =$ ..... **[3]**

25 The functions f, g and h are defined as follows:

$f(x) = 2x + 3$, $g(x) = x^2 - 4$, $h(x) = 5^x$

(a) Find $h(-1)$ as a fraction in its simplest form.

..... **[1]**

(b) Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(d) Find the value of x when $h(x) = 125$.

$x =$ ..... **[1]**

26 Two functions, f and g, are defined as follows:

$$f(x) = \frac{5}{x-2} \qquad g(x) = x + 2$$

(a) State the value of x that must be excluded from the domain of $f(x)$.

$x =$ ..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[4]**

27 $f(x) = 2x - 3$, $g(x) = x^2 - x - 7$

(a) Work out $g(-2)$.

..... **[2]**

(b) Find $gf(x)$ in its simplest form.

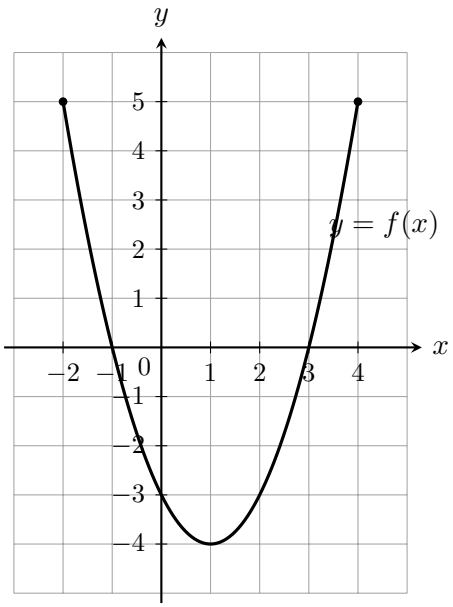
..... **[3]**

(c) Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[3]**

28 The diagram shows the graph of $y = f(x)$ for $-2 \leq x \leq 4$, where $f(x) = x^2 - 2x - 3$.
The function g is defined by $g(x) = 2x + 1$.



(a) Use the graph to write down the range of f for $-2 \leq x \leq 4$.

..... **[2]**

(b) Find $fg(3)$.

$fg(3) =$ ..... **[2]**

(c) Find $gf(x)$, giving your answer in the form $ax^2 + bx + c$.

$gf(x) =$ ..... **[2]**

(d) Solve the equation $g^{-1}(x) = 6$.

$x =$ ..... **[2]**

29 Let $f(x) = \dfrac{2x+1}{3}$ and $g(x) = \dfrac{5}{x-2}$, where $x \neq 2$.

(a) Find $gf(x)$.
Give your answer as a single fraction in its simplest form.

..... **[3]**

(b) Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

(c) Solve the equation

$$f(x) = g(3)$$

$x =$ ..... **[3]**

30 $f(x) = 2x - 1$
$g(x) = x^2 - 4$
$h(x) = \dfrac{2x+1}{x-3}, \quad x \neq 3$

(a) Work out $g(-3)$.

..... **[1]**

(b) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

(c) Find $gf(x)$ in the form $ax^2 + bx + c$.

$gf(x) =$ ..... **[3]**

(d) Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[3]**

31 Consider the functions $f(x) = \dfrac{2x+3}{x-1}$ for $xe1$, and $h(x) = x^2 - 4$ for $x \geq 0$.

Show that $ff(x) = x$ for all $xe1$.

[3]

32 The functions f, g and h are defined as $f(x) = 4 - 3x$, $g(x) = \dfrac{5}{x-2}$, $x \neq 2$, $h(x) = x^2 - x - 7$.

(a) **(i)** Find the value of $f(0.5)$.

..... **[1]**

(ii) Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[2]**

(b) Find the range of $h(x)$.

..... **[2]**

(c) Solve the equation.

$h(x) = 2x - 1$ Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

33 Functions f, g and h are defined as follows.

$f(x) = 3x - 2$

$g(x) = x^2 - 4x$

$h(x) = \dfrac{2x+1}{x-1}$ for $x \neq 1$

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

(b) Find an expression for $g(f(x))$ in the form $ax^2 + bx + c$.

..... **[3]**

(c) Solve the equation $f(g(x)) = x$.
Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... , x =

..... **[4]**

34 Let $f(x) = 3x + 2$, $g(x) = x^2 - x$ and $h(x) = 5^x$.

(a) Find $fg(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[2]**

(b) Find $gf(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[3]**

(c) Solve the equation

$$gf(x) = 0$$

$x =$ ..... , $x =$ ..... **[3]**

(d) Find the value of x when $h(x) = \frac{1}{125}$.

$x =$ ..... **[2]**

35 $f(x) = \dfrac{3}{x+2}, \quad x \neq -2$
$g(x) = 5x - 4$

(a) Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[2]**

(b) Find $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[2]**

(c) Find $ff(x)$ as a single fraction in its simplest form.

$ff(x) =$ ..... **[3]**

(d) Solve the equation $f(x) = g(x)$.
Show all your working.

$x =$ ..... , $x =$ ..... **[4]**

36 Let $f(x) = \dfrac{2x+5}{x-3}$ for $x \neq 3$, and $g(x) = 3x - 1$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Express $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

(c) Solve the equation $gf(x) = x$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[5]**

37 Consider the functions $f(x) = \dfrac{3}{x+2}$, $g(x) = (3x+5)^2$ and $h(x) = \dfrac{3x+5}{x-2}$.

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

(b) Find $fg(x)$.
Give your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

(c) Solve the equation.

$h(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

38 The functions f and g are defined as:

$$f(x) = \frac{2x+3}{x-1} \qquad g(x) = x^2 - 4x$$

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(b) Find $ff(x)$, giving your answer as a single fraction in its simplest form.

$ff(x) =$ ..... **[4]**

(c) Solve the equation $g(x) = f(2)$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

39 The functions f and g are defined as $f(x) = \dfrac{x+5}{x-2}$, $g(x) = x^2 + 3x$.

(a) State the value of x for which $f(x)$ is undefined.

$x =$ ..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(c) Solve the equation

$g(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

(d) Solve the equation

$f(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

1 **(a)** $g(3) = 3^2 + 1 = 10$ **[M1]**

$f(10) = 2(10) - 5 = 15$ **[A1]**

(b) $y = 2x - 5$ OR $x = 2y - 5$ **[M1]**

$\frac{x+5}{2}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** 11 **[B1]**

(b) $\{-7, -4, 2, 11\}$ **[B1]**

(c) $x = 3y - 4$ or $y + 4 = 3x$ **[M1]**

$\frac{x+4}{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** 10 **[B1]**

(b) 5 **[B1]**

(c) $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** **(i)** -14 **[B1]**

(ii) $f(2) = 3(2) - 2 = 4$ **[M1]**

20 **[A1]**

(b) $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** -13 **[B1]**

(b) $y = 4x - 5$ or $x = 4y - 5$ **[M1]**

$\frac{x+5}{4}$ **[A1]**

(c) $4\left(\frac{x}{3} + 2\right) - 5$ **[M1]**

$\frac{4x}{3} + 3$ or $\frac{4x+9}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 18 **[B1]**

(b) $y = 5x - 2$ or $x = \dfrac{y+2}{5}$ **[M1]**

$\dfrac{x+2}{5}$ **[A1]**

(c) $g(2) = 2^2 + 3 = 7$ **[M1]**

$f(7) = 5(7) - 2 = 33$ **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** 7 **[B1]**

(b) Let $y = 4x - 5$ leading to $y + 5 = 4x$ or equivalent **[M1]**

$\dfrac{x+5}{4}$ **[A1]**

(c) $g(2) = 6$ **[M1]**

19 **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** 10 **[B1]**

(b) $y = 3x - 2 \implies 3x = y + 2$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

(c) $f(2) = 3(2) - 2 = 4$ **[M1]**

$g(4) = 4^2 + 4 = 20$ **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** -19 **[B1]**

(b) $f(2) = 4(2) - 7 = 1$ **[M1]**

$g(1) = 2(1)^2 + 1 = 3$ **[A1]**

(c) $y = 4x - 7$ rearranged to $4x = y + 7$ or $x = 4y - 7$ **[M1]**

$\dfrac{x+7}{4}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** 13 **[B1]**

(b) Finds $f(2) = 1$ **[M1]**

-2 **[A1]**

(c) Sets $y = 5 - 2x$ and rearranges to $2x = 5 - y$ **[M1]**

$\frac{5-x}{2}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** 5 **[B1]**

(b) $y = 4x - 3 \implies x = \frac{y+3}{4}$ or $x = 4y - 3$ **[M1]**

$\frac{x+3}{4}$ **[A1]**

(c) $f(1) = 4(1) - 3 = 1$ **[M1]**

$g(1) = 2 - 5(1) = -3$ **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** Calculates $g(3) = 2(3)^2 + 1 = 19$ **[B1]**

Substitutes their 19 into $f(x)$: $4(19) - 3 = 73$ **[B1]**

(b) Let $y = 4x - 3$ and attempts to rearrange for x (e.g., $y + 3 = 4x$) **[M1]**

$\frac{x+3}{4}$ **[A1]**

(c) Sets $4x - 3 = 17$ **[M1]**

$x = 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** 10 **[B1]**

(b) Sets $y = 3x - 2$ or $x = 3y - 2$ OR clearly shows inverse operations $(x + 2) \div 3$ **[M1]**

$\frac{x+2}{3}$ **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $(3x - 2)^2 + 1$ **[M1]**

Correct expansion of $(3x - 2)^2$: $9x^2 - 6x - 6x + 4$ OR $9x^2 - 12x + 4$ **[M1]**

$9x^2 - 12x + 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** $\frac{5}{(2x+1)-3}$ **[M1]**

$\frac{5}{2x-2}$ or $\frac{5}{2(x-1)}$ **[A1]**

(b) $y(x - 3) = 5$ or $xy - 3y = 5$ **[M1]**

$\frac{5}{x} + 3$ or $\frac{5+3x}{x}$ **[A1]**

(c) $2\left(\dfrac{5}{x-3}\right)+1=6$ or $\dfrac{10}{x-3}=5$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 6]*

15 (a) $y(x-1)=3x+2$ **[M1]**

$xy-3x=y+2$ or $x(y-3)=y+2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

(b) $\dfrac{3x+2}{x-1}=x-2$ **[M1]**

$3x+2=(x-1)(x-2)$ leading to $3x+2=x^2-3x+2$ **[M1]**

$x^2-6x=0$ or $x(x-6)=0$ **[M1]**

0 and 6 **[A1]**

[Back to Question] *[Total Marks: 7]*

16 (a) 1 **[B1]**

(b) $y=2x+3 \implies 2x=y-3$ **[M1]**

$\dfrac{x-3}{2}$ **[A1]**

(c) $hk(x)=\dfrac{12}{(2x+3)-1}$ **[M1]**

$\dfrac{12}{2x+2}=\dfrac{6}{x+1}$ **[A1]**

(d) $k(2)=2(2)+3=7$ **[B1]**

$\dfrac{12}{x-1}=7 \implies 12=7(x-1)$ **[M1]**

$12=7x-7 \implies 19=7x \implies x=\dfrac{19}{7}$ (or $2\dfrac{5}{7}$) **[A1]**

[Back to Question] *[Total Marks: 8]*

17 (a) Sets $y=\dfrac{4}{2x-3}$ and multiplies by denominator to obtain $y(2x-3)=4$ **[M1]**

Expands and rearranges to collect terms in x: $2xy=4+3y$ **[M1]**

$\dfrac{4+3x}{2x}$ **[A1]**

(b) Finds $g(-3)=2$ **[B1]**

Equates and attempts to solve: $\dfrac{4}{2x-3}=2 \implies 4x-6=4$ **[M1]**

2.5 (or $\dfrac{5}{2}$) **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $\dfrac{4}{2x-3}+5$ **[M1]**

Expresses over a common denominator: $\dfrac{4+5(2x-3)}{2x-3}$ **[M1]**

$\dfrac{10x-11}{2x-3}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

18 (a) Let $y=5-3x$ leading to $3x=5-y$ or equivalent **[M1]**

$\dfrac{5-x}{3}$ **[A1]**

(b) $x^2-2=34$

$x^2=36$ [M1]

$x=6, x=-6$ **[A1]**

(c) $5-3\left(\dfrac{x-3}{2}\right)$ **[M1]**

$\dfrac{10-3(x-3)}{2}$ or equivalent common denominator step **[M1]**

$\dfrac{19-3x}{2}$ **[A1]**

(d) $h(2)=5-3(2)=-1$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 9]*

19 (a) $y=\dfrac{x+3}{x-1} \implies xy-y=x+3$ **[M1]**

$xy-x=y+3 \implies x(y-1)=y+3$ **[M1]**

$\dfrac{x+3}{x-1}$ **[A1]**

(b) $\dfrac{\dfrac{x+3}{x-1}+3}{\dfrac{x+3}{x-1}-1}$ **[M1]**

$\dfrac{\dfrac{x+3+3(x-1)}{x-1}}{\dfrac{x+3-(x-1)}{x-1}}$ (or equivalent correct manipulation using a common denominator) **[M1]**

$\dfrac{x+3+3x-3}{x+3-x+1}=\dfrac{4x}{4}=x$ with no mathematical errors seen **[A1]**

(c) $f(3)=\dfrac{3+3}{3-1}=3$ **[B1]**

$2x^2-5=3$ **[M1]**

$2x^2=8 \implies x^2=4$ **[M1]**

$2, -2$ **[A1]**

[Back to Question] *[Total Marks: 10]*

20 **(a)** $y(x-1) = 3x + 2$ or $x(y-1) = 3y + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y-3) = y + 2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

(b) $fg(x) = \dfrac{3\left(\dfrac{2}{x+1}\right) + 2}{\dfrac{2}{x+1} - 1}$ **[M1]**

Numerator becomes $\dfrac{6 + 2(x+1)}{x+1}$ or Denominator becomes $\dfrac{2 - (x+1)}{x+1}$ **[M1]**

$\dfrac{2x+8}{1-x}$ **[A1]**

(c) $\dfrac{3x+2}{x-1} = \dfrac{x+2}{x-3} \implies (3x+2)(x-3) = (x+2)(x-1)$ **[M1]**

$3x^2 - 7x - 6 = x^2 + x - 2$ **[M1]**

$2x^2 - 8x - 4 = 0$ or $x^2 - 4x - 2 = 0$ **[M1]**

$2 \pm \sqrt{6}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

21 **(a)** $y(x-1) = 3x + 5$ or $xy - y = 3x + 5$ **[M1]**

$xy - 3x = y + 5$ **[M1]**

$x(y-3) = y + 5$ **[M1]**

$\dfrac{x+5}{x-3}$ **[A1]**

(b) $\dfrac{3x+5}{x-1} = x$ **[M1]**

$3x + 5 = x^2 - x$ **[M1]**

$x^2 - 4x - 5 = 0$ **[M1]**

5 and -1 **[A1]**

(c) $f(2) = \dfrac{3(2)+5}{2-1} = 11$ **[M1]**

$11^2 + 1 = 122$ **[A1]**

[Back to Question] *[Total Marks: 10]*

22 **(a)** **(i)** $\dfrac{4}{(2x+3)-1}$ **[M1]**

$\dfrac{2}{x+1}$ **[A1]**

(ii) $\dfrac{4}{(x^2-4)-1} - 1$ **[M1]**

$\dfrac{4-(x^2-5)}{x^2-5}$ **[M1]**

$\dfrac{9-x^2}{x^2-5}$ **[A1]**

(b) $y(x-1) = 4$ or $x - 1 = \dfrac{4}{y}$ **[M1]**

$xy = 4 + y$ or $x = \dfrac{4}{y} + 1$ **[M1]**

$\dfrac{x+4}{x}$ **[A1]**

(c) $2\left(\dfrac{4}{x-1}\right) + 3 = 5$ **[M1]**

$\dfrac{8}{x-1} = 2$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 11]*

23 **(a)** $y = \dfrac{2x+1}{3} \implies 3y = 2x + 1$ **[M1]**

$\dfrac{3x-1}{2}$ **[A1]**

(b) $hk(x) = \dfrac{2(5-x)+1}{3}$ **[M1]**

$\dfrac{11-2x}{3}$ **[A1]**

(c) $\dfrac{2x+1}{3} = 5 - x$ **[M1]**

$2x + 1 = 15 - 3x \implies 5x = 14$ **[M1]**

2.8 (or $\dfrac{14}{5}$) **[A1]**

[Back to Question] *[Total Marks: 7]*

24 **(a)** $h(5) = \dfrac{4}{5-3} = 2$ **[M1]**

$k(2) = 2(2) + 1 = 5$ **[A1]**

(b) $y(x-3) = 4$ OR $x(y-3) = 4$ **[M1]**

$\dfrac{4}{x} + 3$ OR $\dfrac{4+3x}{x}$ **[A1]**

(c) $h(2x+1) = \dfrac{4}{(2x+1)-3}$ **[M1]**

$\dfrac{4}{2x-2} = 2 \implies 4 = 4x - 4$ **[M1]**

$x = 2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

25 **(a)** $\frac{1}{5}$ **[B1]**

(b) $(2x+3)^2 - 4$ **[M1]**

$4x^2 + 6x + 6x + 9 - 4$ **[M1]**

$4x^2 + 12x + 5$ **[A1]**

(c) $y = 2x + 3 \implies y - 3 = 2x$ **[M1]**

$\frac{x-3}{2}$ **[A1]**

(d) 3 **[B1]**

[Back to Question] *[Total Marks: 7]*

26 **(a)** 2 **[B1]**

(b) Sets $y = \frac{5}{x-2}$ and multiplies by $(x-2)$: $y(x-2) = 5$ **[M1]**

Rearranges to isolate x: $x - 2 = \frac{5}{y} \implies x = \frac{5}{y} + 2$ OR $xy - 2y = 5 \implies x = \frac{5+2y}{y}$ **[M1]**

$\frac{5}{x} + 2$ OR $\frac{5+2x}{x}$ **[A1]**

(c) Equates functions: $\frac{5}{x-2} = x + 2$ **[M1]**

Multiplies by $(x-2)$ to form a quadratic: $5 = (x+2)(x-2)$ **[M1]**

Expands and simplifies to $x^2 = 9$ or $x^2 - 9 = 0$ **[A1]**

3 and -3 **[A1]**

[Back to Question] *[Total Marks: 8]*

27 **(a)** $(-2)^2 - (-2) - 7$ **[M1]**

-1 **[A1]**

(b) $(2x-3)^2 - (2x-3) - 7$ **[M1]**

$4x^2 - 12x + 9 - 2x + 3 - 7$ **[M1]**

$4x^2 - 14x + 5$ **[A1]**

(c) $2x - 3 = x^2 - x - 7$ **[M1]**

$x^2 - 3x - 4 = 0$ or $(x-4)(x+1) = 0$ **[M1]**

$x = 4$ and $x = -1$ **[A1]**

[Back to Question] *[Total Marks: 8]*

28 **(a)** -4 and 5 seen **[B1]**

$-4 \leq f(x) \leq 5$ **[B1]**

(b) $g(3) = 7$ **[B1]**

$f(7) = 32$ **[B1]**

(c) $2(x^2 - 2x - 3) + 1$ **[M1]**

$2x^2 - 4x - 5$ **[A1]**

(d) $x = g(6)$ or $\dfrac{x-1}{2} = 6$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 8]*

29 **(a)** Substitution of $f(x)$ into $g(x)$: $\dfrac{5}{\frac{2x+1}{3} - 2}$ **[M1]**

Writing the denominator as a single fraction: $\dfrac{2x+1-6}{3}$ or $\dfrac{2x-5}{3}$ **[M1]**

$\dfrac{15}{2x-5}$ **[A1]**

(b) Let $y = \dfrac{5}{x-2}$ and attempting to multiply by $(x-2)$ or swapping x and y **[M1]**

$x - 2 = \dfrac{5}{y}$ or $xy - 2y = 5$ **[M1]**

$\dfrac{5}{x} + 2$ or $\dfrac{5+2x}{x}$ **[A1]**

(c) Evaluates $g(3) = 5$ **[B1]**

Equating $f(x)$ to their $g(3)$: $\dfrac{2x+1}{3} = 5$ **[M1]**

$x = 7$ **[A1]**

[Back to Question] *[Total Marks: 9]*

30 **(a)** 5 **[B1]**

(b) $y(x-3) = 2x + 1$ **[M1]**

$xy - 3y = 2x + 1$
$xy - 2x = 3y + 1$
$x(y-2) = 3y + 1$ **[M1]**

$\dfrac{3x+1}{x-2}$ **[A1]**

(c) $(2x-1)^2 - 4$ **[M1]**

$4x^2 - 2x - 2x + 1 - 4$ **[M1]**

$4x^2 - 4x - 3$ **[A1]**

(d) $2x - 1 = x^2 - 4$

$x^2 - 2x - 3 = 0$ **[M1]**

$(x-3)(x+1) = 0$ **[M1]**

$x = 3, x = -1$ **[A1]**

[Back to Question] *[Total Marks: 10]*

31 Substitutes $f(x)$ into f: $ff(x) = \dfrac{2\left(\dfrac{2x+3}{x-1}\right)+3}{\left(\dfrac{2x+3}{x-1}\right)-1}$ **[M1]**

Multiplies numerator and denominator by $(x-1)$: $\dfrac{2(2x+3)+3(x-1)}{(2x+3)-(x-1)}$ **[M1]**

Simplifies to $\dfrac{7x+3}{x+4}$ — note: checks algebra carefully: $4x+6+3x-3 = 7x+3$, $2x+3-x+1 = x+4$; wait: let us verify $f(x) = \dfrac{2x+3}{x-2}$ is self-inverse! With $f(x) = \dfrac{2x+3}{x-1}$, $ff(x) = \dfrac{7x+3}{x+4}$.

[Back to Question] *[Total Marks: 7]*

32 **(a)** **(i)** 2.5 **[B1]**

(ii) $g(4-3x) = \dfrac{5}{(4-3x)-2}$ **[M1]**

$\dfrac{5}{2-3x}$ **[A1]**

(b) Minimum x-coordinate is $-\dfrac{-1}{2(1)} = 0.5$ or via completing square $(x-0.5)^2 - 7.25$ **[M1]**

$h(x) \geq -7.25$ **[A1]**

(c) $x^2 - x - 7 = 2x - 1$ leading to $x^2 - 3x - 6 = 0$ **[M1]**

$\dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-6)}}{2(1)}$ **[M2]**

If M2 not scored, allow M1 for $\dfrac{3 \pm \sqrt{33}}{2}$ or correct use of formula with one error **[M1]**

-1.37 and 4.37 **[A1]**

[Back to Question] *[Total Marks: 9]*

33 **(a)** $y(x-1) = 2x+1 \implies xy - y = 2x+1$ **[M1]**

$xy - 2x = y + 1 \implies x(y-2) = y+1$ **[M1]**

$\dfrac{x+1}{x-2}$ **[A1]**

(b) $g(3x-2) = (3x-2)^2 - 4(3x-2)$ **[M1]**

$9x^2 - 12x + 4 - 12x + 8$ **[M1]**

$9x^2 - 24x + 12$ **[A1]**

(c) $3(x^2 - 4x) - 2 = x$ **[M1]**

$3x^2 - 13x - 2 = 0$ **[A1]**

$x = \dfrac{13 \pm \sqrt{(-13)^2 - 4(3)(-2)}}{2(3)}$ **[M1]**

$x = 4.48$ or $x = -0.15$ **[A1]**

[Back to Question] *[Total Marks: 10]*

34 **(a)** Substitution of $g(x)$ into $f(x)$: $3(x^2 - x) + 2$ **[M1]**

$3x^2 - 3x + 2$ **[A1]**

(b) Substitution of $f(x)$ into $g(x)$: $(3x + 2)^2 - (3x + 2)$ **[M1]**

Correct expansion of $(3x + 2)^2$ to $9x^2 + 12x + 4$ **[M1]**

$9x^2 + 9x + 2$ **[A1]**

(c) Sets their three-term quadratic $gf(x)$ equal to 0: $9x^2 + 9x + 2 = 0$ **[M1]**

Valid method to solve quadratic (e.g., $(3x + 1)(3x + 2) = 0$ or correct use of formula) **[M1]**

$x = -\dfrac{1}{3}$ and $x = -\dfrac{2}{3}$ **[A1]**

(d) $5^x = 5^{-3}$ **[M1]**

$x = -3$ **[A1]**

[Back to Question] *[Total Marks: 10]*

35 **(a)** $x = 5y - 4$ or $y = 5x - 4$ **[M1]**

$\dfrac{x + 4}{5}$ **[A1]**

(b) $\dfrac{3}{(5x - 4) + 2}$ **[M1]**

$\dfrac{3}{5x - 2}$ **[A1]**

(c) $\dfrac{3}{\dfrac{3}{x + 2} + 2}$ **[M1]**

$\dfrac{3}{\dfrac{3 + 2(x + 2)}{x + 2}}$ **[M1]**

$\dfrac{3(x + 2)}{2x + 7}$ **[A1]**

(d) $\dfrac{3}{x + 2} = 5x - 4$

$3 = (5x - 4)(x + 2)$ **[M1]**

$3 = 5x^2 + 10x - 4x - 8$

$5x^2 + 6x - 11 = 0$ **[M1]**

$(5x+11)(x-1)=0$ or correct use of quadratic formula **[M1]**

$x=1, x=-2.2$ (or $-\frac{11}{5}$) **[A1]**

[Back to Question] *[Total Marks: 11]*

36 **(a)** Sets $y=\frac{2x+5}{x-3}$ and rearranges to get $y(x-3)=2x+5$ **[M1]**

Expands and factorises to collect terms in x: $xy-2x=3y+5 \implies x(y-2)=3y+5$ **[M1]**

$\frac{3x+5}{x-2}$ **[A1]**

(b) Substitutes $g(x)$ into $f(x)$: $\frac{2(3x-1)+5}{(3x-1)-3}$ **[M1]**

Expands and simplifies the numerator to $6x+3$ **[M1]**

$\frac{6x+3}{3x-4}$ **[A1]**

(c) Sets up the equation $3\left(\frac{2x+5}{x-3}\right)-1=x$ **[M1]**

Eliminates the denominator to obtain $6x+15-(x-3)=x(x-3)$ (or equivalent) **[M1]**

Simplifies to form the correct 3-term quadratic equation: $x^2-8x-18=0$ **[A1]**

Correct use of the quadratic formula $x=\frac{-(-8)\pm\sqrt{(-8)^2-4(1)(-18)}}{2(1)}$ **[M1]**

9.83 and -1.83 **[A1]**

[Back to Question] *[Total Marks: 11]*

37 **(a)** $y=\frac{3x+5}{x-2} \implies y(x-2)=3x+5$ **[M1]**

$xy-2y=3x+5 \implies xy-3x=2y+5 \implies x(y-3)=2y+5$ **[M1]**

$\frac{2x+5}{x-3}$ **[A1]**

(b) $\frac{3}{(3x+5)^2+2}$ **[M1]**

$\frac{3}{9x^2+30x+25+2}=\frac{3}{9x^2+30x+27}$ **[M1]**

$\frac{3}{3(3x^2+10x+9)}=\frac{1}{3x^2+10x+9}$ **[A1]**

(c) $\frac{3x+5}{x-2}=x \implies 3x+5=x(x-2)$ **[M1]**

$3x+5=x^2-2x \implies x^2-5x-5=0$ **[A1]**

$\frac{-(-5)\pm\sqrt{(-5)^2-4(1)(-5)}}{2(1)}$ **[M1]**

$\frac{5\pm\sqrt{25+20}}{2}=\frac{5\pm\sqrt{45}}{2}$ **[M1]**

5.85 and -0.85 **[A1]**

[Back to Question] *[Total Marks: 11]*

38 **(a)** Sets $y = \dfrac{2x+3}{x-1}$ and multiplies: $y(x-1) = 2x+3$ **[M1]**

Expands bracket and collects x terms on one side: $xy - 2x = y + 3$ **[M1]**

Factorises x: $x(y-2) = y+3$ **[M1]**

$\dfrac{x+3}{x-2}$ **[A1]**

(b) Substitutes $f(x)$ into itself: $\dfrac{2\left(\dfrac{2x+3}{x-1}\right)+3}{\left(\dfrac{2x+3}{x-1}\right)-1}$ **[M1]**

Forms common denominators for numerator and denominator: $\dfrac{\dfrac{2(2x+3)+3(x-1)}{x-1}}{\dfrac{(2x+3)-(x-1)}{x-1}}$ **[M1]**

Correctly simplified numerator: $4x+6+3x-3 = 7x+3$ OR correctly simplified denominator: $2x+3-x+1 = x+4$ **[B1]**

$\dfrac{7x+3}{x+4}$ **[A1]**

(c) Calculates $f(2) = \dfrac{2(2)+3}{2-1} = 7$ **[B1]**

Forms correct quadratic equation: $x^2 - 4x - 7 = 0$ **[B1]**

Correct substitution into quadratic formula: $\dfrac{4 \pm \sqrt{(-4)^2 - 4(1)(-7)}}{2}$ OR $\dfrac{4 \pm \sqrt{44}}{2}$ **[M1]**

5.32 and -1.32 **[A1]**

[Back to Question] *[Total Marks: 12]*

39 **(a)** 2 **[B1]**

(b) Let $y = \dfrac{x+5}{x-2}$ **[M1]**

$y(x-2) = x+5$

xy - 2y = x + 5[M1]

$xy - x = 2y + 5$

x(y - 1) = 2y + 5[M1]

$\dfrac{2x+5}{x-1}$ or equivalent **[A1]**

(c) $x^2 + 3x - 10 = 0$ **[M1]**

$(x+5)(x-2) = 0$ or correct substitution into quadratic formula **[M1]**

$x = -5, x = 2$ **[A1]**

(d)

$$\frac{x+5}{x-2} = x$$

$\mathrm{x}^2 - 2x = x + 5$ [M1]

$x^2 - 3x - 5 = 0$ **[M1]**

$x = \dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-5)}}{2}$ **[M1]**

$x = 4.19, x = -1.19$ **[A1]**

[Back to Question] *[Total Marks: 12]*